INCH-POUND
A-A-59724D
July 12, 2022
SUPERSEDING
A-A-59724C
October 19, 2010

COMMERCIAL ITEM DESCRIPTION

TRUCK, FIRE FIGHTING (QUINT)

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers a commercial Quint. It has a 4x2 or 6x4 chassis with a diesel engine and an automatic transmission, a cab for at least five seated persons with at least four doors, a 1,250-gallon-per-minute (gpm) fire pump, a 500-gallon minimum capacity water tank, hose storage areas, a 75-foot aerial ladder with a permanently mounted waterway, and equipment compartments. The Quint is intended to combat structural and similar type fires.

2. SALIENT CHARACTERISTICS. The Quint shall be in accordance with the applicable requirements of the following chapters of National Fire Protection Association (NFPA) 1901, *Standard for Automotive Fire Apparatus* (Current Edition).

Chapter	Title
1	Administration
2	Referenced Publications
4	General Requirements
9	Quint Fire Apparatus
12	Chassis and Vehicle Components
13	Low-Voltage Electrical Systems and Warning Devices
14	Driving and Crew Area
15	Body, Compartments, and Equipment Mounting
16	Fire Pump and Associated Equipment
19	Water Tanks
20	Aerial Devices
23	Line Voltage Electrical Systems

2.1 Administration.

2.1.1 Environmental conditions.

Beneficial comments, recommendations, additions, deletions, clarifications, and any other data that may improve this document should be sent to: AFLCMC/EZSS, 2145 Monahan Way, Area B, Bldg 28, Wright-Patterson AFB, OH 45433 or email engineering.standards@us.af.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

FSC 4210

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2.1.1.1 Altitude. Unless otherwise specified (see 6.2), the vehicle, including the pumping system, shall be designed for operation at or below 2,000 feet above sea level.

2.1.1.2 Temperature range. The vehicle shall be capable of satisfactory storage and operation in temperatures ranging from 0 to 125°F. The vehicle shall be equipped with a cab, chassis, and firefighting agent winterization system, permitting full operation at 0°F. The winterization system shall not detract from the performance of the vehicle or the firefighting system in ambient temperatures up to 125°F. The vehicle chassis winterization system shall maintain the engine coolant, lubricants, fuel, and electrical systems operational at ambient temperatures of 0°F. The vehicle agent winterization shall provide protection of the agent pump, agent piping system(s), firefighting systems, and water tank to 0°F. The vehicle agent-winterization system shall provide sufficient insulation and heating capacity, by means of either hot circulating liquids, immersion heaters, and/or forced-air heat exchangers, to permit satisfactory operation of the vehicle and firefighting systems for a 2-hour period at ambient temperatures as low as 0°F with the vehicle fully operational and the engine running. At the end of this 2-hour period, the vehicle shall be capable of successfully discharging its agents from all discharges. All compartments not winterized shall be marked on the interior of the compartment as not winterized and not to be used for storage of items subject to freezing. While in operation, the appropriate systems requiring winterization shall be powered by the vehicle. While in storage, the winterization system shall be powered through the electrical shoreline connection (see 2.11.2).

2.1.2 Foreign object damage. All loose metal parts, such as pins and valve caps, shall be securely attached to the vehicle with wire ropes or chains. "Dog tag" style beaded chains shall not be provided. Removable panels, if provided, shall be attached with captive fasteners.

2.1.3 Roadability.

2.1.3.1 Operating terrain. The vehicle shall operate on paved and graded gravel roads.

2.1.3.2 Acceleration. The fully loaded vehicle shall accelerate from 0 to 35 miles per hour (mph) within 25 seconds on a level, paved road.

2.1.3.3 Maximum speed. The fully loaded vehicle shall attain a minimum top speed of 50 mph on a level, paved road.

2.1.3.4 Gradeability. The fully loaded vehicle shall be able to maintain a speed of at least 20 mph while ascending any paved slope up to and including 6.0 percent. The fully loaded vehicle shall also be able to maintain a speed of at least 5.0 mph while ascending any paved slope up to and including 30 percent.

2.1.4 Overall dimensions. Overall dimensions shall be the minimum consistent with the operational performance and the design constraints necessary to achieve the specified performance. Overall dimensions shall not exceed the following:

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Length	450 inches
Width	102 inches (excluding mirrors)
Height	144 inches

2.1.5 Turning diameter. The fully loaded vehicle shall have a wall-to-wall turning diameter of less than three times the length of the vehicle.

2.1.6 Angles of approach and departure. The fully loaded vehicle shall have angles of approach and departure of not less than 8 degrees.

2.1.7 Vehicle stability. Vehicle shall comply with rollover stability criteria as defined by NFPA 1901.

2.1.8 Manuals.

2.1.8.1 Technical manuals. The overall format for the manuals may be military specification, commercial, or a combination of both. Each technical manual shall have a title page.

a. The contractor shall validate the technical manuals for accuracy prior to submission to the procuring activity for verification. The contractor shall submit one complete set to the procuring activity for verification 30 days before the first production test. Any changes or corrections noted by the procuring activity shall be corrected, and updated pages or manuals shall be submitted to the procuring activity.

b. Once the technical manuals have been approved by the procuring activity and a technical order (TO) number is assigned, the contractor shall pack two complete sets of both hard copy and compact disc (CD) with each vehicle. One complete set of manuals in rewriteable CD format shall be sent to the procuring activity for stock. The procuring activity's address will be provided.

c. The contractor shall grant the United States (U.S.) Air Force a non-exclusive, non-assignable, royalty free U.S. Government (Government) Purpose License to scan into CENTRA (the U.S. Air Force repository) and to reproduce and distribute (either electronically or via hard medium) copies or facsimiles of manuals produced and distributed by the contractor for this CID. These rights extend to Government agencies only, and the data contained in the technical manuals is not to be sold, disclosed, or otherwise provided to any other entity or entities outside the Government. The license shall remain in effect as long as the vehicles described in the required technical manuals remain under Government control and usage. In addition, the contractor shall grant permission for the Government to place an Air Force TO number and publication date as well as distribution, warning, handling, and destruction statements on the cover/title pages of the TO.

2.1.8.1.1 Operator's manuals. The operator's manual shall include all information required for the safe and efficient operation of the vehicle, including the fire-extinguishing equipment, and any special attachments or auxiliary equipment. The operator's manual shall include at least the following:

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a. Full illustrations and descriptions of the location and function of all controls and instruments.

b. Safety information that is consistent with the safety standards established by the Occupational Safety and Health Administration (OSHA).

c. Checks and adjustments in preparation for placing the vehicle for service upon receipt from the contractor.

d. Preparation for shipment or storage.

e. Warranty information and period of the warranty for the complete vehicle and for any component warranty that exceeds the warranty of the complete vehicle. Addresses and telephone numbers shall be provided for all warranty providers.

f. General description of and step-by-step instructions for the operation of the vehicle and its fire extinguishing system(s) and auxiliary equipment.

g. Description of the post-operational procedures (draining, flushing, et cetera).

h. Checklists for the daily maintenance inspection and mission-readiness checks that the operator is expected to perform.

i. Procedures for towing a disabled vehicle.

j. Procedures for changing a tire.

k. Schedules for required preventative maintenance and required periodic maintenance.

l. A line-art drawing of the vehicle, including front, rear, left, and right side views, showing basic dimensions and weights (total vehicle and individual axle weight for the unloaded and fully loaded vehicle).

m. Procedures and maximum speed limits for towing a trailer, including hazards, risks, and special requirements.

2.1.8.1.2 Service manuals. The service manual shall identify any special tools and test equipment required and shall cover troubleshooting and maintenance, as well as minor and major repair procedures. The text shall contain performance specifications, tolerances, and fluid capacities; current, voltage, and resistance data; test procedures; and such illustrations and exploded views as may be required to permit proper maintenance by qualified mechanics. The manual shall contain an alphabetical subject index, as well as a table of contents. The service manual shall contain at least the following, where applicable:

a. Firefighting system schematic(s).

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b. Hydraulic schematic.

c. Pneumatic schematic.

d. Electrical schematic.

e. Winterization schematic.

f. Schedules for required preventative maintenance and required periodic maintenance.

g. Location, procedure, and interval for parts of the truck and equipment which require lubrication.

2.1.8.1.3 Parts manual. The parts manual shall include illustrations and exploded views, as needed, to properly identify all parts, assemblies, subassemblies, and special equipment. All components of assemblies shown in illustrations or exploded views shall be identified by reference numbers which correspond to the reference numbers in the parts lists. All purchased parts shall be cross-referenced with the original manufacturer's name and part number. The parts manual shall provide the description, length, dimensions, and quantity of each item used per vehicle. The manual shall contain a numerical index. The parts manual shall contain a list of all of the component vendor names, addresses, and telephone numbers referenced in the parts list.

2.1.9 Painting, plating, and corrosion control.

2.1.9.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall not be used in intimate contact with each other.

2.1.9.2 Finish. Exterior surfaces shall be prepared, primed, and painted with polyurethane paint in accordance with all of the paint manufacturer's instructions and recommendations. Unless otherwise specified (see 6.2), the exterior finish color shall be Candy Apple Red, Sikkens Color Number FLNA3021, DuPont Color Number 97902U or 4737U, PPG Color Number 71528 (the PPG name for this color is Cardinal Red), or equal. When specified, the exterior finish color shall be Desert Sand, Color Number 30313 of SAE AMS-STD-595. When specified, the exterior finish color shall be Olive Drab with Flat Finish, Color Number 34064 of SAE AMS-STD-595. The interior of all compartments shall be painted with an impact-resistant, textured coating that resists stains, scuffs, chips, and scratches.

a. For vehicles painted Candy Apple Red, the cab upper body (from the bottom of the windshield) and roof shall be painted White, Color Number 17875 of SAE AMS-STD-595. All bright metal and anodized parts, such as mirrors, horns, light bezels, treadplate, and roll-up compartment doors, shall not be painted. Compartment interiors shall be painted with an impact-resistant, textured coating that resists stains, scuffs, chips and scratches.

b. For vehicles painted Desert Sand or Olive Drab, all exterior surfaces, including all normally bright metal and anodized parts and any interior surfaces visible with any compartment door open (but not the interior of the cab), shall be painted body color. This includes fenders, compartment doors, shelves, and mounting hardware, but does not include items mounted in the compartments. Non-metallic materials may be black or gray.

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2.1.9.3 Horizontal reflective striping. Horizontal reflective striping in accordance with NFPA 1901 shall be applied around the vehicle in an approximate plane with the headlights. Bright metal trim or anodized parts may interrupt the reflective stripes. The pattern shall be 10 inches wide with three reflective stripes (one inch reflective, one inch body color, six inches reflective, one inch body color, and one inch reflective). The reflective stripes shall be white.

2.1.9.4 Chevron striping. In accordance with NFPA 1901 at least 50 percent of the rear-facing vertical surfaces, visible from the rear of the apparatus, excluding any pump panel areas not covered by a door, shall be equipped with retroreflective striping in a chevron pattern sloping downward and away from the centerline of the vehicle at an angle of 45 degrees. 3M part number 983-17 (Red) and 983-23 (Florescent Yellow/Green) reflective diamond grade or equivalent shall be utilized. Each stripe in the chevron shall be a single color alternating between red and fluorescent yellow in accordance with NFPA 1901. Each stripe shall be 6 inches (150 millimeters) in width in accordance with NFPA 1901 and the example provided below.



2.1.9.5 Lettering. Vehicles painted Candy Apple Red shall have the letters "UNITED STATES" and "AIR FORCE" or any other applicable military service applied in synthetic or encapsulated gold leaf, with outline and black shadow, on the front door on both sides in long-radius elliptical arches above and below the lettering center line. The size of the lettering shall be a minimum of 2½ inches to a maximum of 6 inches. The Air Force Fire Emergency Services logo shall be placed at a location to be determined during the post award meeting. Vehicles painted Desert Sand or Olive Drab shall have no lettering or logo.

2.1.9.6 Aerial Ladder Lettering. Vehicles painted Candy Apple Red shall also have synthetic or encapsulated gold leaf letters "U.S. AIR FORCE" with outline and black shadow located on both side panels of the retracted ladder. Base names are not authorized. Vehicles painted Desert Sand or Olive Drab shall have no lettering.

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2.1.10 Identification plate. A permanently marked identification plate shall be securely mounted at the driver's compartment. The identification plate shall contain the following information:

a. Nomenclature.
b. Manufacturer’s make and model.
c. Manufacturer’s serial number.
d. U.S. Air Force vehicle registration number.
e. National Stock Number (NSN).
f. Vehicle curb weight: pounds (kilograms).
g. Payload, maximum: pounds (kilograms).
h. Gross Vehicle Weight (GVW): pounds (kilograms).
i. Fuel and capacity and type: gallons (liters).
j. Date of delivery (month and year).
k. Warranty (months and miles (kilometers)).
l. Contract number.
m. Order number.
n. Paint color and number.
o. Length, width, and height of vehicle in inches and centimeters.

2.1.11 Delivery. All units shall be driven to the receiving installation for CONUS locations or port of call for OCONUS locations.

2.1.12. Warranty. The warranty period shall not begin until training has been completed at the receiving installation.

2.1.13. Air Force registration numbers. The manufacturer shall maintain a list of Air Force registration numbers and corresponding serial numbers for the purpose of identifying vehicles affected by service bulletins.

2.2 General requirements.

2.2.1 Hose storage.

2.2.1.1 Hose storage area. The vehicle shall have a hose bed, with removable cover and two adjustable dividers, capable of carrying 1,000 feet of 5-inch hose and 800 feet of 3-inch hose. Hose bed covers will be red for vehicles painted Candy Apple Red, tan for Desert Sand vehicles, and dark green for Olive Drab vehicles. The hose storage area shall be equipped with a hose-retention device to prevent inadvertent hose deployment. Covers will be secured by snap fasteners. Hose beds shall be of a prismatic design. Narrowing that requires hose and coupling to negotiate turns shall not be allowed. Only one hose divider shall be required.

2.2.1.2 Cross lay hose beds. The vehicle shall have three cross-lay hose beds with one or more removable covers. Two hose beds shall be capable of carrying 200 feet of 1¾-inch pre-connected, double-jacketed hose each. The third hose bed shall be capable of carrying 200 feet of 3-inch double-jacketed hose. Sufficient clearance shall be provided between the hose and cover to allow a pre-connected hose nozzle to be pulled through from either side of the vehicle. Cross-lay hose

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bed covers will be red for vehicles painted Candy Apple Red, tan for Desert Sand vehicles, and dark green for Olive Drab vehicles. Covers will be secured by snap fasteners.

2.2.1.3 Hose-board trays. The vehicle shall be equipped with one drop-in running board hose/equipment tray located under discharge panel on officer's side. The tray shall be equipped with a mat and shall be designed to allow for proper drainage. The pump panel side shall have a slide out step.

2.2.2 Suction hose.

2.2.2.1 Hard suction hose. Two 10-foot long, lightweight 6-inch hard suction hoses, with 6-inch National Hose thread long-handle female couplers on one end and 6-inch National Hose thread long-handle male couplers on the other, shall be mounted on the vehicle above the compartments on one side. A suction strainer shall also be mounted on the vehicle. The hose and suction strainer mounting system design shall be subject to approval by the procuring activity.

2.2.2.2 Soft suction hose. The vehicle shall be equipped with a 20-foot long, 5-inch soft suction hose in accordance with NFPA 1961, with a 4½-inch long-handle National Hose thread female coupler on one end and a 6-inch long handle National Hose thread female coupler on the opposite end. The couplers shall be in accordance with NFPA 1963. The soft suction hose and couplers shall be stored in a compartment between the truck cab and the extended front bumper.

2.2.3 Ground ladders. Ground ladders shall be provided as listed in NFPA 1901 and shall be of aluminum construction. The ground ladders shall be mounted in locations readily accessible to personnel standing on the ground. One 16-foot straight ladder with hooks shall be mounted to the side of the aerial ladder for ease of operations. Each ladder shall be individually accessible; it shall not be necessary to remove or relocate one ladder to access another ladder. The ground-ladder mounting system design shall be subject to approval by the procuring activity.

2.2.3.1 Ground ladder testing. The manufacturer shall perform service testing of all ground ladders in accordance with NFPA 1932 prior to vehicle acceptance and shall provide certification and testing completion in both document and PDF format.

2.2.4 Minor equipment.

2.2.4.1 Fire hose and nozzles. The vehicle shall be designed to accommodate all items listed in NFPA 1901. The fire hose and nozzles shall not be provided with the vehicle. The vehicle shall have compartment space for the full complement of equipment, listed in NFPA 1901. The self-contained breathing apparatus (SCBA), the spare SCBA cylinders, and the Automatic External Defibrillator shall be provided by the purchaser. Individual compartment storage shall be included for five purchaser-provided one-hour-capacity SCBA air bottles**.**

2.2.4.2 Chocks. Two folding wheel chocks and mounting hardware shall be provided as loose equipment. The wheel chocks shall hold the apparatus, when loaded to its GVWR or GCWR, on a hard surface with a 20-percent grade with the transmission in neutral and the parking brake released.

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2.2.5 Backboard storage. The vehicle shall have storage of a long spine backboard. The location shall be determined at the post-award conference.

2.3 Chassis and vehicle components.

2.3.1 Engine. The truck shall have a diesel engine that is certified to comply with the Environmental Protection Agency (EPA) on-highway emission requirements at the time of manufacture. The engine shall run on Ultra-Low-Sulfur Diesel Fuel unless otherwise specified (see 6.2).

2.3.1.1 Fuel filters. Primary and secondary fuel filters and a fuel/water separator shall be provided. The fuel/water separator shall include a water coalescer and a drain valve, and shall be in accordance with SAE J1839. A combination fuel filter and fuel/water separator may be provided. Fuel-filter elements shall be replaceable without loss of engine prime.

2.3.2 Exhaust system. The exhaust system outlet(s) shall be directed away from personnel accessing equipment compartments.

2.3.2.1 Vehicle-mounted diesel-exhaust filter system. A vehicle-mounted, direct-source capture, exhaust filter system shall be supplied with the vehicle that shall prevent exposure to, and contamination from, exhaust emissions in addition to the manufacturer's after-treatment device (ATD) in accordance with NFPA 1500. This system shall be installed after the engine manufacturer's ATD and before the diffuser tip in the end-tailpipe. All vehicles without a DEF/DPF system shall be supplied with a direct-source capture system that includes an electronic control module which engages a diverter unit, to direct diesel exhaust from the vehicle's exhaust system into a ceramic filter where diesel particulates and gasses are trapped. All apparatuses/vehicles equipped with this system must include a spare filter upon delivery of the apparatus/vehicle to the end user. This filter system shall work automatically whenever the vehicle exits and returns to the station. In addition, the system shall have the capability to be used while on-scene, outside of the station. The diesel-exhaust removal system shall travel with the vehicle. The system shall not require building modifications or hanging hoses for the system to operate. The installation of the system shall be performed by the exhaust-removal system manufacturer and shall not be subcontracted. The vehicle-mounted exhaust filter system shall meet all NFPA, National Institute for Occupational Safety and Health (NIOSH), and OSHA standards for preventing exposure to carcinogenic compounds that exist in diesel exhaust.

2.3.3 Brake system. The vehicle shall be equipped with an all-wheel antilock brake system; the brakes shall be fully air-actuated. Brakes shall be in accordance with Code of Federal Regulations (CFR) 49 CFR 393.40 through 393.42(b), 393.43, and 393.43 through 393.52. The braking system complete with all necessary components shall include the following:

a. Air compressor having a capacity of not less than 12 standard cubic feet per minute (scfm).

b. Air storage reservoir(s), each tank equipped with drain and with safety and check valves between the compressor and the reservoir tank.

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c. Automatic moisture ejector on air storage reservoir.

d. Automatic slack adjusters on cam brakes or internal self-adjusting brakes on wedge and disc brakes on all axles.

e. Spring-set parking brakes.

All components of the braking system shall be installed in such a manner as to provide adequate road clearance when traveling over uneven or rough terrain, including objects liable to strike and cause damage to the brake-system components. No part of the braking system shall extend below the bottom of wheel rims, to ensure, in case of a flat tire, that the weight of the vehicle will be supported by the rim and the flat tire and not be imposed on any component of the braking system. Slack adjusters and air chambers shall be located above the bottom edge of the axle carrier.

2.3.3.1 Air dryer. A replaceable-cartridge desiccant air dryer shall be installed in the air-brake system. The dryer shall have the capability of removing not less than 95 percent of the moisture in the air being dried. The dryer shall have a pre-cooler and a filter to screen out oil and solid contaminants. The dryer shall have an automatic self-cleaning cycle and a thermostatically controlled heater to prevent icing of the purge valve.

2.3.3.2 Compressed-air shoreline. A flush-mounted, checked, auto-eject compressed-air shoreline connection shall be provided to maintain brake-system pressure while the vehicle is not running. It shall be located on the exterior of the vehicle, either adjacent to the driver's door or within 6 inches of the left side front corner of the body. An air chuck shall be provided as loose equipment.

2.3.3.3 Auxiliary braking system. The service brakes shall be augmented by one of the following auxiliary braking systems:

a. A system which opens all or some of the engine exhaust valves near the end of the compression stroke, thereby converting vehicle motion to a pumping loss. The engine brake shall be approved by the engine manufacturer.

b. A hydrodynamic retarder integral with the transmission.

c. A controlled gate valve in the exhaust manifold, which produces engine back pressure. The exhaust brake shall be approved by the engine manufacturer.

d. A dash-mounted switch shall be provided to activate, modulate, or cut out the brake augmentation. The switch shall be marked to indicate its position. When active, the system shall be fully controlled by means of the conventional driving controls to apply retardation during vehicle deceleration, and to cut it out in the other operating modes.

2.3.4 Tires and wheels. The vehicle shall be equipped with tubeless steel-belted radial tires with highway-type tread rated for specified highway speeds as well as off-road conditions, mounted on steel-disc wheel assemblies. If all tire and wheel assemblies are identical, one spare tire and wheel assembly shall be provided. If two different tire and wheel assemblies are provided, two spare tire and wheel assemblies shall be provided, one of each configuration. Spare tires and wheels are not

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required to be mounted on the vehicle. The spare tire(s) and rim(s) shall be shipped with the vehicle as loose equipment.

2.3.5 Tire chains. When specified (see 6.2), the vehicle shall be equipped with driver-activated automatic-type tire chains for the rear axles.

2.3.6 Fenders. Rear fenders and fender liners having tire-chain clearance shall be provided. For vehicles painted Desert Sand or Olive Drab, the fender flares shall be painted the job color.

2.3.7 Steering. The vehicle shall be equipped with power steering.

2.3.8 Transmission. A fully automatic transmission with a hydraulic torque converter and at least four forward speeds shall be provided. The normal driving-range selector position shall provide at least four gear ratios without movement of the selector. The net torque capacity and the net power rating of the transmission shall exceed the output ratings of the engine. The transmission shall have a direct-drive lockup for pumping.

2.3.9 Fuel tank. Fuel tank(s) having a minimum usable capacity in accordance with NFPA 1901 or 50 gallons, whichever is greater, shall be provided. The location of the fuel tank(s) shall protect it (them) from mechanical damage during normal use of the vehicle. The tank fill location or fuel tank cap shall be marked as to the type of fuel the vehicle shall be serviced with.

2.3.10 License-plate bracket. A lighted license-plate bracket shall be provided at the front bumper and the left rear of the vehicle.

2.4 Low-voltage electrical systems. The vehicle shall have a 12-volt electrical and starting system.

2.4.1 Alternator. A single alternator charging system in accordance with NFPA 1901 shall be provided. The minimum continuous electrical load shall include operation of the air-conditioning system.

2.4.2 Batteries. Batteries shall be of the maintenance-free type. The addition of water shall not be required during the battery's normal service life. The battery cover and vent system shall be designed to prevent electrolyte loss during service and to keep the top of the battery free from electrolyte.

2.4.2.1 Battery compartment. The batteries shall be enclosed in a ventilated, corrosion-resistant, weather-resistant box or compartment and shall be readily accessible.

2.4.3 Battery charger or conditioner. The vehicle shall have a direct current (DC) taper-type battery charger or an automatic battery conditioner, providing a minimum 15-amp output. The charger/conditioner shall be permanently mounted on the vehicle in a properly ventilated, accessible location. The charger/conditioner shall be powered from the electrical shoreline receptacle (see 2.11.2). A charging indicator shall be installed next to the receptacle. When a battery conditioner is provided, it shall monitor the battery state of charge and, as necessary, automatically charge or maintain the batteries without gassing, depleting fluid level, overheating, or overcharging.

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2.4.4 Warning lights. All warning lights shall use light-emitting diode (LED) elements. The warning-light system, related components, and devices shall be in accordance with NFPA 1901.

2.4.4.1 Light bar. A six-element LED light bar, or two three-element bars, with both forward- and side-facing heads, shall be mounted on the cab roof. Forward-facing lenses shall be red-white-red-red-white-red, with the white lights switched off in blocking right-of-way mode. Rearward-facing lenses shall be red-amber-red-red-amber-red. The light bar shall be separately switched from the warning-light panel. All trucks shall be equipped with a steady red front warning light as required by California state law regardless of shipping destination.

2.4.4.2 Warning light color. When specified (see 6.2), the rearward, red lights shall be replaced with amber. When specified (see 6.2), all red warning lights shall be replaced with blue.

2.4.4.3 Headlight flashing system. A high-beam, alternating/flashing, headlight system shall be provided. The headlight flasher shall be separately switched from the warning-light panel.

2.4.5 Audible warning devices.

2.4.5.1 Siren. The vehicle shall be equipped with an electronic siren system. The amplifier unit shall include volume control and selection of "Radio," "PA," "Manual," "Yelp," "Wail," and "Hi-Lo" (European) modes, and a magnetic noise-canceling microphone. The amplifier, microphone, and controls shall be within reach of the driver and right front passenger. Siren-activating foot switches shall be located in front of the driver and the right front passenger. The siren speaker shall be rated at 100 watts minimum and shall be located in a guarded position in the front bumper.

2.4.5.2 Horn. Dual forward-facing air horns shall be installed in protected locations near the front of the truck. The air horn shall be activated by foot switches located in front of the driver, the officer, and a switch at pump panel.

2.4.6 Work lighting.

2.4.6.1 Cab interior lights. Cab interior light levels shall be sufficient for reading maps or manuals.

2.4.6.2 Compartment lights. White LED lighting sufficient to provide an average minimum illumination of 2.0 foot-candle shall be provided in each compartment greater than 4.0 cubic feet and having an opening greater than 144 square inches. Where a shelf is provided, this illumination shall be provided both above and below the shelf. Lights shall automatically illuminate only when the respective doors are opened.

2.4.6.3 Ladder, step, and area lights. Non-glare white LED lighting shall be provided at ladders and access steps where personnel work or climb during night operations. These area lights shall be controlled with switches on the cab instrument panel and near the light sources.

2.4.6.4 Spotlights. Three LED spotlights rated at 12,000 lumens minimum each shall be mounted on the cab roof, one center brow, one over the driver’s door, and one over the officer’s door. The spotlights shall be controlled from inside the cab.

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2.4.7 Scene lights. The vehicle shall be equipped with four 12-volt, 12,000-lumen minimum scene lights, two on each side, mounted at the top front and rear corners of the body. Switches for the scene lights shall be located in both the pump panel areas and on the cab instrument panel.

2.4.8 Radio circuit. The vehicle shall have two separate 30-amp circuits, with breakers and at least 6 feet of wire routed to a space provided adjacent to the driver and officer for purchaser provided radios and other electrical equipment. The wiring shall be tagged indicating its purpose.

2.4.9 Radio Antennas. An antenna mounting base with coaxial cable and waterproof cap shall be provided for 2-way radios. The mount shall be located on the cab roof to the rear of the front seats and will be clear of any obstructions. The cable shall be routed to the center console, length of the cable shall be determined at post-award.

2.4.10 Intercom system. A three-way intercom system shall be installed to provide communications between the turntable control station, pump operator's panel and the aerial tip. The intercom system shall include three speakers and three control modules; one with a push-to-talk button at the turntable control station, one with a push-to-talk button at the pump operator's panel and one hands-free at the aerial tip. The control modules shall have push-button volume control and a LED volume display. The hands-free module shall constantly transmit to the other module unless the push-to-talk button is pressed. The intercom shall have active noise cancellation and be designed for exterior use.

2.4.11 Auxiliary power outlet. The vehicle shall be equipped with one 12-volt auxiliary power outlet near each passenger/crew member seated position for portable electronic equipment.

2.5 Driving and crew areas.

2.5.1 Cab. The vehicle shall have a tilting or non-tilting cab with at least four doors with seating for at least five persons. The cab shall provide a minimum vertical head clearance of 47 inches, as measured from the H-point. A manual override provision for the cab hydraulics shall be provided as a backup. The cab shall have a body constructed entirely of non-corroding materials, which is certified for fire service, and able to pass the relevant tests in SAE J2422, SAE J2420. The cab shall have a wrap-around windshield and a matching-contour cab face. Cab door openings shall extend for the full vertical height of the side panels. Steps and handrails shall be provided for all crew doors, and at least one grab handle shall be provided for each crew member, located inside the cab for use while the vehicle is in motion. The crew cab area shall be fitted with a lockable medical supply cabinet. The medical cabinet shall be centered between two rear-facing seats. The medical cabinet opening shall face the rear of the apparatus. The cabinet shall be provided with 2 adjustable shelves. The width and depth of the cabinet shall be the maximum possible without hindering the movement and comfort of the firefighters with full gear in the cab. A vehicle-data recorder shall be provided in accordance with NFPA 1901. Provisions shall be provided for fire-helmet storage for all seated positions in accordance with NFPA 1901.

2.5.1.1 Seats. The truck shall have seats in accordance with NFPA 1901. The driver's seat shall be of an adjustable air-suspension design. The crew member seats shall be of a non-suspension design. Each crew member seat (excluding the drive position) shall have a backrest, brackets, and

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pad covers designed to store a one-hour capacity SCBA. All SCBA-type seats in the cab shall have a hands-free auto-clamp style bracket in its backrest. The bracket shall include an automatic spring clamp that allows the occupant to store the SCBA bottle by pushing it into the seat back. In the event of an accident, the inertial components within the clamp shall constrain the SCBA bottle in the seat and shall exceed the NFPA standard of 9G. Bracket designs shall not have manual restraints (belts, straps, buckles) that could be inadvertently left unlocked and allow the SCBA to move freely within the cab during an accident. Each seat shall be provided with a Type 2 seatbelt assembly (i.e., 3-point restraint) in accordance with 49 CFR 571.209. All seats shall be equipped with a seatbelt sensor in compliance with NFPA 1901. All seatbelts shall be red in color.

2.5.1.2 Cab interior sound level. The maximum sound level at any seat location shall not exceed 90 dBA without any warning devices in operation, as measured in accordance with 49 CFR 393.94(c), "Vehicular interior noise levels test procedure," except that the test shall be performed with the vehicle traveling at a steady speed of 45 mph on a level, hard, smooth surface road.

2.5.1.3 Windshield and windows. The windshield and windows shall be of tinted safety glass.

2.5.1.4 Heated Mirrors. Combination flat and convex outside rearview mirrors shall be installed on each side of the cab, mounted on fold-back west-coast style brackets. The flat mirrors shall be of the motorized remote-control type, providing not less than 60 degrees horizontal rotational viewing range. The flat mirrors shall also have electrically heated heads. Mirror remote and heating controls shall be located on the instrument panel within reach of the seated driver.

2.5.1.5 Climate-control system. The manufacturer’s standard heater/defroster system shall be provided. The manufacturer’s standard air-conditioning system shall also be provided. In 100ºF ambient temperature with 50-percent relative humidity and at maximum compressor speed, the air conditioning system shall cool the fully occupied cab to 75ºF within 30 minutes.

2.5.1.6 Instruments and controls. Gauges shall be provided for oil pressure, coolant temperature, and automatic-transmission temperature. All switches and/or controls that activate systems or system components shall be labeled as to their function. In addition to the instruments and controls required by 14.3.6 of NFPA 1901, the following shall be provided within convenient reach of the seated driver:

a. Master warning-light control switch,

b. Work-light switch(es), and

c. Compartment "Door Open" warning light and intermittent alarm that sounds when a compartment door is open and the parking brakes are released or the transmission is in any position other than neutral.

d. An engine-hour meter.

2.5.2 Back-up camera. A hard-wired, rear-vision camera and in-cab monitor shall be provided to aid the driver in safely backing up the vehicle. The monitor shall be located to the right side of the driver. A switch shall be provided to allow the driver to manually activate the back-up camera

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from within the cab. The back-up camera shall also be switched "on" automatically whenever the vehicle is in the reverse mode of operation.

2.5.3 Active air-purification system. The vehicle shall be equipped with an active air-purification system to provide purification of the air and surfaces inside the cab. The system shall contain a capability to generate hydrogen peroxide (H_2O_2) and reactive oxygen species to sanitize against various microbial species. The system shall produce H_2O_2 at a minimum of 0.02 parts per million (ppm) to sanitize the atmosphere inside the apparatus. The H_2O_2 concentration shall not exceed OSHA recommendations. The system shall be sized per application to support virus and bacteria kill rates. The system shall be stand-alone and contain its own airflow mechanism, with a stainless-steel outer housing. The system shall be environmentally friendly and not emit direct UVC outside of the unit. The system shall be resistant to dust particles normally found in apparatus. The working temperature of the system shall be -22 to 149 ºF (-30 to 65 ºC). The system shall be permitted to operate at any time with or without occupants in the cab and shall pose no harm to the occupants from H_2O_2, ozone, or UVC light. The system shall be powered and operating at all times, both in the fire station and during all vehicle operations.

2.5.3.1 Air-purification system certification. The air-purification system shall be third-party tested to verify H_2O_2 production at 0.02 ppm and to support virus and bacteria kill rates. The manufacturer must be ISO 9001:2015 certified and an EPA-registered establishment. The system shall meet all applicable sections of IEC 61373:2010 for shock and vibration and SAE J1455 for electrical specifications. The third-party testing must be performed for effectiveness against SARS-CoV-2 in a chamber at least 1,280 cubic feet in size.

2.6 Body, compartments, and equipment mounting.

2.6.1 Body. The vehicle shall have a body constructed entirely of non-corroding materials. The body shall be at least as strong as if it were constructed of 0.125-inch thick aluminum. A copy of the fire-apparatus manufacturer's warranty shall be included with the bid. The warranty shall state that the body shall be free of structural failures caused by defective design or workmanship for a warranty period of fifteen (15) years from the date the new vehicle is first delivered or 100,000 miles, whichever occurs first, and that defective parts under the warranty shall be repaired or replaced without charge to the original purchaser.

2.6.2 Compartments. The vehicle body shall include sufficient compartment space to hold all of the equipment listed in 2.2.4 or a minimum of 145 cubic feet in addition to ladder storage. The compartment space at the left and right side of the rear compartments shall extend to the maximum length possible without exceeding the depth of the tailboard.

2.6.2.1 Compartment doors. Compartments shall have clear anodized aluminum, counterbalanced, non-locking, roll-up doors and will be provided with an aluminum trey/guard that protects the roll-up door from damage. Guards shall not be provided on medical cabinets located inside cabs. Door-latch handles shall be full-width bar type. Door straps shall be provided to assist in closing the compartment doors when the rolled-up door height exceeds six feet above the ground.

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2.6.2.2 Scuffplates. Replaceable scuffplates shall be provided to prevent body damage from sliding equipment in and out of the compartments. The scuffplates shall be attached in a manner that does not allow them to become loose during normal use.

2.6.2.3 Drip rails. Drip rails shall be provided over each compartment door. If the drip rails are not integral with the body, they shall be of anodized or extruded aluminum and shall have a bright finish for trucks painted Candy Apple Red.

2.6.2.4 Shelves. An adjustable and removable compartment shelf shall be provided for every 18 inches of each vertical storage-compartment door opening. Shelving adjustments shall require no more than common hand tools, and shall not require disassembly of fasteners. Shelves shall support a minimum of 500 pounds without permanent deformation. Each shelf shall be accessible to crew members standing on the ground. A pull-out and tip-down configuration shall be used for shelving over 54 inches from the ground when the compartment is more than 18 inches deep. Each shelf shall have ¼-inch diameter drain holes located in each corner of the shelves so as to allow for drainage of any water from the stowed equipment.

2.6.2.5 Drainage mats. Each compartment floor and shelf shall be covered with a removable mat designed to allow for drainage of any water from the stowed equipment.

2.6.3 SCBA storage tubes. Tubes for storage of five SCBA one-hour bottles shall be installed adjacent to the wheel wells for two SCBA cylinders on each side of the body. The tubes shall be in accordance with NFPA 1901.

2.6.4 Ladder, handrails, and walkways. Ladders, stepping, standing, and walking surfaces shall be in accordance with NFPA 1901. Handrails shall be provided in accordance with NFPA 1901.

2.6.5 Body and compartment lighting. All standard body and compartment lighting shall be LED type and provided in accordance with NFPA 1901.

2.7 Fire pump and associated equipment.

2.7.1 Fire pump. The fire pump shall be a single-stage centrifugal design, with a thick-walled cast-iron pump body and a bronze impeller, capable of providing 1,250 gpm of water at 150 pounds per-square-inch gauge (psig). The pump shall be driven from the chassis propulsion engine through a split-shaft power takeoff with ratings to handle the full torque and power applied. The cast-iron pump body must not contribute in any way to the premature failure of the fire pump.

2.7.2 Piping and associated components. All metallic surfaces of the piping and associated components that come into contact with the water shall be of passivated stainless steel or bronze and shall be capable of storing brackish/saltwater. The discharge piping shall flow water at a minimum of 1,250 gpm.

2.7.3 Intake connections. The vehicle shall have three 6-inch intake connections: one on each side and one at the rear of the vehicle, directly connected to the aerial waterway and equipped with a 6-inch, 45-degree, turn-down fitting. The top of the rear intake connection shall be no higher than 36 inches above the ground; it shall be equipped with at least 4-inch nominal diameter piping. The

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6-inch intakes on the sides of the vehicle shall be equipped with short stub connections. Six-inch quarter-turn valves shall be provided for the 6-inch side intakes. The valves shall have National Hose threads and shall be equipped with a pressure-relief device in accordance with NFPA 1901. The vehicle shall have two 2½-inch intake connections, one on each side, adjacent to the 6-inch intake connections and fitted with 45-degree turn-down fittings. Each intake connection shall be gated and shall have National Hose threads.

2.7.4 Discharge connections. The vehicle shall be equipped with a standard master-discharge connection on the right side. It shall be provided with no less than 4-inch full-flow piping and valve with a 4-inch NST to 5-inch Storz adapter on a 30-degree turn-down elbow. The vehicle shall be equipped with two 2½-inch discharges on driver's side and a 3-inch pipe and full-flow valve on officer's side. Each 2½-inch discharge connection shall be equipped with no less than 2½-inch full-flow piping and valve with a 2½-inch adapter and a 45-degree turn-down fitting. The vehicle shall be equipped with three swivel discharges to accommodate the pre-connected cross-lay hoses. Two shall be plumbed with no less than 2-inch piping or high-pressure hose, have 1½-inch swivel connections, and flow at least 200 gpm through 200 feet of 1¾-inch fire hose. The third shall be plumbed with no less than 2½-inch piping or high-pressure hose, have a 2½-inch swivel connection, and flow at least 250 gpm through 200 feet of 3-inch fire hose. All discharge connections shall have National Hose threads.

2.7.4.1 Monitor. The vehicle shall be equipped with a single 1,000-gpm monitor nozzle permanently mounted at the top of the aerial ladder as a pinnable waterway. The horizontal and vertical position of the monitor, as well as the discharge pattern, shall be controlled by individual switches located at both the turntable and the tip of the ladder. A fog tip and stacking-type tips with stream straighteners, 2-inch, 1¾-inch, 1½-inch, and 1⅜-inch shall be furnished.

2.7.5 Pump-operator control panel. A pump-operator control panel in accordance with NFPA 1901 shall be provided. It shall be mounted on the left-center side of the vehicle. Analog gauges shall be provided. The panel shall have master intake and discharge pressure gauges, not less than 4½-inch diameter, and individual discharge outlet pressure gauges, not less than 2-inch diameter. The panel shall also have a flowmeter for the aerial waterway, the standard master discharge connection, and one of the 2½-inch discharge connections on the left side. Each discharge outlet shall be controlled from the pump operator's panel through a manual valve having locking features. The water-tank level indicator shall be of the five-light type; that is, it shall have lights to show the water tank level at quarter-tank-capacity intervals and have a flashing "refill" or "tank empty" light. Discharge and intake controls shall be color coded in accordance with NFPA 1901. All components shall be labeled.

2.7.6 Extended front bumper. The vehicle shall be equipped with an extended front bumper. The bumper shall have a center tray, with a secured cover, and 1½ inch outlet. The center tray shall be in accordance with NFPA 1901. The outlet connection shall be located inside the center tray

2.8 Water tank. The vehicle shall have a water tank with a certified capacity of at least 500 gallons.

2.8.1 Water-tank material. The water tank shall be constructed of polypropylene.

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2.8.2 Water-tank baffling. The water tank shall be designed for dynamic baffling in accordance with NFPA 1901.

2.8.3 Tank-to-pump intake line. All metallic surfaces of the tank-to-pump intake line and associated components that come into contact with the water shall be of passivated stainless steel or bronze and shall be capable of storing brackish/saltwater. The tank-to-pump intake line shall be capable of delivering water at a flow rate in accordance with NFPA 1901 for the certified capacity of the water tank provided. The tank-to-pump valve(s) shall be air or electrically actuated with a manual override.

2.9 Aerial device. The vehicle shall be equipped with a telescoping aerial ladder. The rated vertical height of the ladder shall be at least 75 feet when measured in accordance with NFPA 1901. The maximum elevation of the ladder should not exceed 75 degrees (nominal). The rated capacity of the aerial ladder shall be a minimum load of 500 pounds (114 kg) carried at the tip of the fly section with the aerial ladder at 90 degrees to the horizontal and at maximum extension, while discharging water through the full range of monitor or nozzle movements as permitted by the aerial manufacturer. The aerial device shall be equipped with a permanently installed waterway.

2.10 Foam-proportioning system. When specified (see 6.2), the vehicle shall be equipped with an electronic, variable-speed, fully automatic, discharge-side, direct-injection foam-proportioning system for both Class A and Class B foam concentrates. It shall be in accordance with NFPA 1901. With 0.5-percent concentration of Class A foam, the system shall properly proportion foam for flow rates of up to and including 1,000 gpm. The system shall also accommodate 1-percent Class A foam concentrate as well as 3-percent and 6-percent Class B foam concentrates. Tanks with a minimum usable capacity of 25 gallons each shall be provided for both Class A and Class B foam concentrates. Each foam-concentrate tank shall be provided with a 1½-inch female foam-tank fill connection, with a rocker-lug plug and a replaceable ¼-inch mesh strainer, on the left side of the vehicle in a location readily accessible from the ground. Each fill connection shall be provided with a check valve to prevent loss of foam concentrate when connection or disconnection is made. Foam tanks shall be equipped with a manual shut-off valve accessible from the outside of the vehicle.

2.11 Line voltage electrical system.

2.11.1 Auxiliary generator. A minimum 15-kilowatt (kW) (continuous rating), 120/240-volt alternating current (VAC), 60-Hertz (Hz), hydraulically driven generator shall be provided. The generator shall be activated from the driver's position and shall have a monitor light to indicate operation. A governor shall regulate engine speed to match the generator output to the connected load. Gauges shall monitor the operation of the generator system and indicate the connected load. Access for maintenance shall be provided above the generator. Individual circuit-breaker tripping or failure shall not affect operation of other active circuits.

2.11.2 Electrical shoreline connection. The battery charger/conditioner shall be powered from a covered, three-wire, straight-blade, polarized, insulated, labeled, recessed, 120-volt, NEMA 5-20P male plug inlet, in a weatherproof alternating current (AC) auto-eject receptacle. It shall be located on the exterior of the vehicle, either adjacent to the driver's door or within 6 inches of the left-side front corner of the body. The shoreline connection’s location shall not pose a tripping hazard to

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pedestrians walking by the vehicle when the shore line is connected. A weatherproof charge meter shall be installed next to the receptacle. A 50-foot long, three-wire, 20-amp rated, 120-volt, AC power cable, with straight blade (non-twist-lock), NEMA 5-20R and 5-20P style, in accordance with NEMA WD-6, connectors, shall be provided. When equipped with a winterization system, the winterization system shall have a labeled activation switch with a pilot light. When specified (see 6.2), the receptacle(s) and power cable(s) shall be for 220 volts in lieu of 110 volts.

2.11.3 Distribution box. A weatherproof distribution box shall be located adjacent to the pump operator's control panel. It shall have a minimum of seven 120-volt, 15-amp circuits, with ground-fault indicator (GFI) breakers, and two 240-volt, 50-amp circuits, with GFI breakers.

2.11.4 Receptacles. Two 120-volt AC, 20-amp, electrical outlets shall be provided, one at each rear corner of the body. Each location shall have weatherproof cover(s) and shall provide one L5-20R and one duplex NEMA 5-20R receptacles. The receptacles shall be powered by the auxiliary generator (see 2.11.1). See example provided below.



2.11.5 Cable reels. The vehicle shall be equipped with one electrical cable reel located in a dunnage area with access through a compartment on either side. The reel shall be equipped with 200 feet of 10/3 SO, 20-amp, 600-volt, 90ºC insulated electrical cable. The electrical cables shall be equipped with rubber ball stops to prevent cable pull through during rewinding operations. A four-way roller guide shall be provided for the cable reel to prevent chafing of cable insulation. The cable reel shall have an electric rewind motor with provisions for manual rewind in the event of motor failure; the manual-rewind handle shall be securely stored near the cable reel. A portable, lighted, weatherproof, GFCI, duplex-outlet box, with built-in circuit breakers and a twist-lock, NEMA L5-20R-style receptacle in accordance with NEMA WD-6, shall be provided for the cable end. The cable reel shall be powered by the auxiliary generator (see 2.11.1).

2.11.6 Aerial-ladder lights. The vehicle shall be equipped with two 1,000-watt equivalent LED lights at the tip of the ladder, powered by the auxiliary generator (see 2.11.1). The lights shall be positioned as to not interfere with egress from aerial device. The lights shall be controlled by switches in the cab and at the turntable.

2.12 Workmanship. The vehicle, including all parts and accessories, shall be fabricated in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, burrs, defects, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of welding, brazing, soldering, riveting, and painting; alignment of parts; tightness of fasteners; etc. Wire ties may be used for bundling vehicle wiring, but not for support

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of wire bundles. Support must be provided by insulated metal clamps. The vehicle shall be thoroughly cleaned of all foreign matter.

2.13 Human factors engineering. All system operations, servicing, and maintenance functions shall be configured to be accomplished by a range of personnel from a 5th-percentile female to a 95th-percentile male.

2.14 Training. A minimum of three days of training shall be provided at the receiving base following the delivery of the vehicle. Training shall consist of driver/operator training, and informational overview training for vehicle-maintenance personnel. One three-day training session shall be required at each location, regardless of the number of vehicles shipped to that location.

3. REGULATORY REQUIREMENTS.

3.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements and promotes economically advantageous lifecycle costs. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with 23.403 of the Federal Acquisition Regulation (FAR). However, used, rebuilt, or refurbished items shall not be provided.

3.2 Green Procurement Program. Green Procurement Program (GPP) is a mandatory federal acquisition program that focuses on the purchase and use of environmentally preferable products and services. GPP requirements apply to all acquisitions using appropriated funds, including services and new requirements. FAR 23.404(b) applies and states the GPP requires 100% of EPA-designated product purchases that are included in the Comprehensive Procurement Guidelines list that contains recovered materials, unless the item cannot be acquired a) competitively within a reasonable timeframe, b) meeting appropriate performance standards, or c) at a reasonable price. The prime contractor is responsible for ensuring that all subcontractors comply with this requirement.

4. PRODUCT CONFORMANCE PROVISIONS

4.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The Government reserves the right to require proof of such conformance.

4.2 Commercial item requirement. The vehicle furnished shall comply with the "commercial item" definition of FAR 2.101 as of the date of award. The Government reserves the right to require the manufacturer to prove that their product complies with the referenced commerciality requirements and each salient characteristic of this CID. The manufacturer shall provide an itemized technical proposal that describes how the proposed model complies with each salient characteristic of this CID**;** a paragraph-by-paragraph response to the salient characteristics section of this CID shall be provided**.** The response to each paragraph shall clearly state “Fully Comply” or “Bid with Exception” in addition to other data the contractor wishes to submit. The proposal shall be provided

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with the pricing submission. Failure to provide this information may deem a vendor as non-responsive, and their proposal may be rejected. The manufacturer shall provide two copies of their commercial descriptive catalogs with their offer as supporting reference to the itemized technical proposal. The manufacturer shall identify all modifications made to their commercial model in order to comply with the requirements herein.

4.3 Inspection requirements.

4.3.1 General inspection requirements. Apparatus used in conjunction with the inspections specified herein shall be laboratory-precision type, calibrated at proper intervals to ensure laboratory accuracy. Calibration certificates shall be available for Government review.

4.3.2 Test rejection criteria. Throughout all tests specified herein, the vehicle shall be closely observed for the following conditions, which shall be cause for rejection:

a. Failure to conform to design or performance requirements specified herein or in the contractor's technical proposal.

b. Any spillage or leakage of any liquid, including fuel, coolant, lubricant, or hydraulic fluid, under any condition, except as allowed herein.

c. Structural failure of any component, including permanent deformation, or evidence of impending failure.

d. Evidence of excessive wear.

e. Interference between the vehicle components or between the vehicle, the ground, and all required obstacles, with the exception of normal contact by the tires.

f. Misalignment of components.

g. Evidence of undesirable roadability characteristics, including instability in handling during cornering, braking, and while traversing all required terrain.

h. Conditions that present a safety hazard to personnel during operation, servicing, or maintenance.

i. Overheating of the engine, transmission, or any other vehicle component.

j. Evidence of corrosion.

k. Failure of the firefighting system.

4.3.3 Detailed inspection requirements.

4.3.3.1 Examination of product. Each vehicle shall be examined to verify compliance with the salient characteristics herein. A contractor-generated checklist that identifies each relevant

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requirement and the inspection results shall be used. Particular attention shall be given to materials, workmanship, dimensions, surface finishes, protective coatings and sealants and their application, welding, fastening, and markings. Proper operation of each vehicle function shall be verified. Each production vehicle shall be inspected to a reduced version of the checklist that has been approved by the procuring activity.

4.3.3.2 Road tests. The following tests shall be performed in accordance with NFPA 1901.

4.3.3.2.1 Maximum speed and acceleration test. The vehicle shall be tested to demonstrate compliance with 2.1.3.2 and 2.1.3.3. For the first-production unit, a time-distance recorder shall be used to record data for this test.

4.3.3.2.2 Gradeability test. The first-production vehicle shall be tested to demonstrate compliance with 2.1.3.4.

4.3.3.2.3 Auxiliary braking system test. The vehicle shall be tested in accordance with NFPA 1901.

4.3.3.2.4 Service brake system test. The vehicle shall be tested in accordance with NFPA 1901. For the first-production unit, a time-distance recorder shall be used to record data for this test.

4.3.3.2.5 Turning diameter test. The fully loaded first-production vehicle shall be tested in accordance with SAE J695 to demonstrate compliance with 2.1.5.

4.3.3.2.6 Roadability test. The fully loaded first-production vehicle shall be driven over 10 miles of paved roads and 5 miles of gravel roads. All loads shall be removed and all structure and surfaces shall be visibly inspected for failure or permanent deformation.

4.3.3.3 Low-voltage electrical system performance tests. The vehicle shall be tested in accordance with NFPA 1901.

4.3.3.4 Cab interior sound-level test. The cab interior sound levels of the first-production vehicle shall be measured in accordance with 49 CFR 393.94(c), "Vehicular interior noise levels test procedure," except that the test shall be performed with the vehicle traveling at a steady speed of 45 mph on a level, hard, smooth surface road.

4.3.3.5 Test of fire pump and associated equipment. The vehicle shall be tested in accordance with NFPA 1901. The costs associated with the independent certification shall be included in the price of the vehicle.

4.3.3.6 Aerial device certification tests. The vehicle shall be tested in accordance with NFPA 1901. The costs associated with the independent certification shall be included in the price of the vehicle.

4.3.3.7 Foam-proportioning system test. If the vehicle is equipped with a foam-proportioning system (see 6.2), it shall be tested in accordance with 2.10 and NFPA 1901.

4.3.3.8 Line voltage electrical system testing. The vehicle shall be tested in accordance with NFPA 1901.

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5. PACKAGING.

5.1 Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Source of documents.

6.1.1 Government documents. Copies of specifications, standards, and handbooks are available online at https://quicksearch.dla.mil/.

6.1.2 Code of Federal Regulations (CFR). CFRs may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402, or online at https://www.ecfr.gov/cgi-bin/ECFR?page=browse.

6.1.3 Environmental Protection Agency (EPA) documents. Copies of EPA documents may be obtained from US EPA National Headquarters Building, 1200 Pennsylvania Avenue NW, Washington, DC 20004, or online at https://www.epa.gov/.

6.1.4 Federal Acquisition Regulations (FAR). Copies of FARs may be obtained from the Superintendent of Documents, Government Printing Office, Washington, DC 20402, or online at https://www.acquisition.gov/?q=browsefar.

6.1.5 International Electrotechnical Commission (IEC) standards. Copies of IEC standards may be obtained from IEC Secretariat, 3 rue de Varemba, PO Box 131, CH-1211 Geneva 20, Switzerland, or online at https://www.iec.ch.

6.1.6 ISO standards. Copies of ISO standards may be obtained from Chemin de Blandonnet 8, CP 401, 1214 Vernier, Geneva, Switzerland, or online at https://www.iso.org/.

6.1.7 National Electrical Manufactures Association (NEMA) documents. Copies of NEMA documents may be obtained from the addressed to National Electrical Manufactures Association, 1300 North 17th Street, Suite 900, Arlington, VA 22209, or online at https://www.nema.org/.

6.1.8 National Fire Protection Association (NFPA) documents. Copies of NFPA documents may be obtained from NFPA, 1 Batterymarch Park, Quincy, MA 02269-9101, or online at https://www.nfpa.org/.

6.1.9 Occupational Safety and Health Association (OSHA) standards. Copies of OSHA standards may be obtained from OSHA, 200 Constitution Avenue, Washington, DC 20210, or online at https://www.osha.gov/.

6.1.10 SAE International (SAE) standards. Copies of SAE standards may be obtained from the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096, or online at https://www.sae.org/.

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6.2 Ordering data. The contract or order should specify the following:

a. Altitude for which vehicle operation is to be designed, if greater than 2,000 feet above sea level (see 2.1.1.1).

b. Finish color required (Desert Sand or Olive Drab, in place of Candy Apple Red) (see 2.1.9.2).

c. Diesel fuel is required in lieu of ultra-low-sulfur diesel fuel (see 2.3.1).

d. If driver-activated automatic-type tire chains are required (see 2.3.5).

e. Emergency warning-light color required (amber or blue in place of red) (see 2.4.4.2) (NOTE: Applies to USAFE ONLY).

f. If a foam-proportioning system is required (see 2.10)

g. Electrical shoreline voltage, 220 volts instead of 110 volts (see 2.11.2). (Note: Applies to some USAFE and South West Asian (CENTAF) locations only).

6.3 Key Words.

Aerial devices
Ground ladders
Hoses
Pumper
Structural type usage
Water tanks

MILITARY INTERESTS:

Custodian:
Air Force - 184

Preparing activity:
Air Force - 184

Agent:
Air Force - 110

Project 4210-2022-007

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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